Seat No.	

Set P

B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination: October/November - 2025
Operating Systems (G06-GE-OE-0301)

Day & Date: Tuesday, 28-10-2025 Max. Marks: 30
Time: 12:00 PM To 01:30 PM

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternative. **06**

1) _____ scheduling algorithm is non-preemptive scheduling.
a) SJF Scheduling b) Round-robin scheduling
c) SRTF Scheduling d) MLFQ scheduling

2) Process synchronization can be done on _____.
a) hardware level
b) software level
c) both hardware and software level
d) Deadlock

3) The operating system where fixed time slot is allocated to each active process is _____.
a) Real time O.S. b) Multiprogramming O.S.
c) Batch O.S. d) Time-sharing O.S.

4) _____ is the disadvantages of the priority scheduling algorithm.
a) it schedules in a very complex manner
b) its schedules take up a lot of time
c) it can lead to some low priority process waiting indefinitely for the CPU
d) All of these

5) The entry of all the PCBs of the current processes is in _____.
a) Program Counter b) Process Register
c) Process Unit d) Process Table

6) The number of processes completed per unit time is known as _____.
a) Output b) Throughput
c) Efficiency d) Capacity

Q.2 Answer the following. (Any Three) **06**

a) Define Scheduler.
b) Define Semaphores.
c) Define Multiprogramming operating System.
d) What is Process Synchronization?
e) What is Operating system?

Q.3 Answer the following. (Any Two) **06**

a) Explain Types of Schedulers.
b) Explain the different Services provided by Operating System.
c) Explain Batch Operating System.

Q.4 Answer the following. (Any Two) **06**

a) Explain Dining Philosophers problem.
b) Explain Process Control Block with diagram.
c) Explain SJF Scheduling algorithms.

Q.5 Answer the following. (Any One) **06**

a) Consider the following set of processes, with the length of CPU- burst time given in milliseconds and priority of the process (consider 1 as highest priority):

Process	Arrival time	Burst time	Priority
P1	0	10	3
P2	1	17	1
P3	3	3	3
P4	4	7	4
P5	5	12	2

Solve the following questions:

i) Draw the 3 Gantt chart illustrating the execution of these process using SJF, priority and Round Robin (Quantum = 5).
ii) What is average waiting time for each process and average turnaround time in each scheduling algorithm?

b) Explain FCFS Scheduling Algorithms in detail.